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▲ SENTINELEDGE

RMM Agent Competitive Analysis & Feature Gap Report

Strategic comparison against industry-leading platforms with prioritized implementation roadmap

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NAVIGATION

Contents

01	Executive Summary	
02	SentinelEdge Current State	Architecture, capabilities, and current feature set
03	Competitor Profiles	NinjaOne, Datto RMM, N-able, Atera, Kaseya VSA, ConnectWise, Pulseway
04	Feature Comparison Matrix	Side-by-side capability comparison across 25 feature areas
05	Gap Analysis	Critical missing features with business impact assessment
06	Implementation Proposals	14 feature implementation blueprints with technical specifications
07	Prioritized Roadmap	Phased implementation timeline (Phase 1–4)
08	Conclusions	

OVERVIEW

Executive Summary

This document presents a comprehensive competitive analysis of SentinelEdge Agent against the leading Remote Monitoring and Management (RMM) platforms in the MSP market as of Q1 2026. The analysis covers NinjaOne, Datto RMM (Kaseya), N-able N-central, Atera, Kaseya VSA, ConnectWise Automate/RMM, and Pulseway.

SentinelEdge Agent has built a solid technical foundation: a lightweight Go-based Windows Service with Vaultwarden-secured credentials, deny-by-default command allowlisting, automatic heartbeat/polling, hardware inventory collection, and GitHub Releases-based auto-update. However, the current feature set covers only a fraction of what enterprise MSP agents deliver.

7	25	14	4
COMPETITORS ANALYZED	FEATURE AREAS ASSESSED	IMPLEMENTATION PROPOSALS	ROADMAP PHASES

STRATEGIC FINDING

The most critical gaps are **real-time threshold alerting**, **patch management automation**, **Windows Event Log monitoring**, and **interactive remote terminal access**. These four capabilities are universally present in all seven competitors and represent the baseline for any commercially viable RMM agent in 2026.

Key Differentiators Competitors Hold

Beyond the baseline gaps, competitors are actively investing in AI-powered autonomous remediation (Atera's Robin, ConnectWise AI scripting), hyperscale patch coverage (ConnectWise: 7,000+ apps, NinjaOne: 200+ third-party), mobile-first management

(Pulseway), native Microsoft 365 management (Datto RMM), and deep EDR/security integration. These represent the next generation of differentiation for SentinelEdge.

OUR FOUNDATION

What SentinelEdge Agent Has Today

The agent is written in Go 1.21+, targeting Windows amd64, and installed as a native Windows Service via `github.com/kardianos/service`. It communicates with the SentinelEdge API (`saapi.ardepa.site`) using authenticated HTTP via resty/v2.

Current Feature Set

Implemented ✓

Capabilities available in the current agent build

- Windows Service lifecycle (Start/Stop/Run)
- YAML configuration via Viper (env var overrides)
- Vaultwarden OAuth2 secrets management
- Agent self-registration with API token storage
- Heartbeat every 300s (configurable)
- Poll-based command queue (every 30s)
- PowerShell command execution (deny-by-default allowlist)
- SHA256 hash + regex command allowlist with RWMutex
- Max 5 concurrent commands (semaphore)
- Hardware inventory via WMI/PowerShell (24h cycle)
- Software inventory (Windows Registry)
- Auto-update from GitHub Releases (SHA256 verified)
- Download URL allowlist enforcement
- Multi-tenant TenantID in all API requests
- Bearer token auth on all requests
- Build-time version injection via ldflags

SECURITY DEBT (FROM AUDIT 2026-04-01)

Two known security items remain open: **F-01 CRITICAL** — TLS certificate pinning not implemented (requires pin distribution infrastructure); **F-05 MEDIUM** — No nonce/timestamp on commands (replay attack vector). Additionally, `Agent.token` is stored in plaintext in Supabase (API-side issue).

MARKET LANDSCAPE

Competitor Profiles

NinjaOne (NinjaRMM)

Enterprise Leader

Gartner Magic Quadrant Leader 2026 · 23 quarters as #1 RMM on G2

The dominant player. Known for fast time-to-value, cross-platform support (Windows, Mac, Linux, VMware, SNMP), and a clean UI. Offers 200+ third-party app patches, five scripting languages, and deep remote access integration.

- Patch Management (OS + 200+ third-party apps)
- 5 scripting languages (PS, JS, Bash, VBScript, Batch)
- Role-Based Access Control (RBAC)
- Managed Antivirus (AV) + EDR integration
- Remote terminal + remote desktop
- Network device monitoring (SNMP)
- Windows / Mac / Linux / VMware agents
- Real-time threshold alerting
- Device approval workflow
- Password vault management
- Backup integration (NinjaBackup)
- MDM for iOS/Android

Datto RMM

MSP Focused

Now part of Kaseya · Unique: native Microsoft 365 management module

The only RMM with a native M365 management module. Emphasizes security-first design with native ransomware detection, mandatory 2FA, and agent encryption. Strong automation policies engine.

- Native ransomware detection
- Microsoft 365 management (native module)
- Automation Policies (pre-defined workflows)
- Mandatory 2FA enforcement
- PowerShell + Bash + Python scripting
- Remote desktop control
- Patch management + vulnerability scanning
- Network discovery (ESXi monitoring)
- ComStore (script/component library)
- Monitoring templates (CPU/disk/network)
- Datto Backup integration
- SNMP network device monitoring

N-able N-central

Enterprise Scale

650+ scripts · Advanced Linux agent (2026 roadmap) · NetPath diagnostics

N-central is the flagship for large MSPs and enterprise IT. Ships with 650+ pre-built scripts, advanced patch management, network path diagnostics, and Cloud Commander for multi-tenant M365 management. 2026 roadmap includes an optimized Linux agent with log visibility and flexible deployment.

- 650+ pre-built automation scripts
- NetPath network diagnostics
- Cloud Commander (multi-tenant M365)
- SSH management for Cisco/Fortinet/HP/Juniper
- Advanced patch management
- EDR and AV add-ons
- Risk Intelligence module
- Automation Manager (drag-and-drop)
- Sensitive path control (Linux)
- 5-minute detection check cycle
- Log visibility per endpoint
- Take Control (remote desktop)

Atera

AI Pioneer

AI-native: Robin autonomous agent + AI Copilot · Per-technician pricing

Atera is the most AI-forward RMM. **Robin** (relaunched March 2026) autonomously resolves complex device, network, and cloud issues without technician intervention. The AI Copilot assists with script generation, ticket summaries, and remote session analysis. All-in-one platform with integrated PSA and billing.

- Robin — autonomous AI agent (self-healing)
- AI Copilot (script gen, ticket summaries)
- Integrated PSA and billing
- Automated patch management
- Splashtop remote access integration
- Network discovery
- SNMP monitoring
- Real-time monitoring with threshold alerts
- AI-powered ticket routing
- Software deployment automation
- Multi-client dashboard
- Billing automation per device

Kaseya VSA + ConnectWise Automate + Pulseway

3 Platforms

Additional major platforms completing the competitive landscape

Kaseya VSA: Agent Procedures engine for complex workflows, vPro/AMT management, IT Glue integration, mobile VSA app. **ConnectWise Automate**: Expanded from 350 to 7,000+ third-party patches (2025), AI-assisted PowerShell drafting, multi-machine simultaneous management, RPA capabilities. **Pulseway**: Best-in-class mobile app, real-time push alerts, visual workflow builder for automation, MDM for iOS/Android — the only RMM with truly mobile-first management.

SIDE-BY-SIDE COMPARISON

Feature Comparison Matrix

● Full support ◖ Partial/Add-on ○ Not available SE SentinelEdge

Feature	SE	Ninja	Datto	N-able	Atera	Kaseya	CW	Pulseway
Windows Service Agent	●	●	●	●	●	●	●	●
Hardware Inventory	●	●	●	●	●	●	●	●
Auto-Update Mechanism	●	●	●	●	●	●	●	●
PS/Script Execution	●	●	●	●	●	●	●	●
Secrets Management (Vault)	●	●	◖	◖	●	●	◖	◖
Command Allowlist	●	◖	◖	◖	◖	◖	◖	◖
Real-Time Threshold Alerting	○	●	●	●	●	●	●	●
Patch Management	○	●	●	●	●	●	●	●
Windows Event Log Monitor	○	●	●	●	●	●	●	●
Remote Terminal/Shell	○	●	●	●	●	●	●	●
Service Monitoring (Win Services)	○	●	●	●	●	●	●	●
Process Monitoring	○	●	●	●	●	●	●	●

Feature	SE	Ninja	Datto	N-able	Atera	Kaseya	CW	Pulseway
Software Deployment (MSI/EXE)	○	●	●	●	●	●	●	●
File Transfer (push/pull)	○	●	●	●	◐	●	●	●
Vulnerability Scanning (CVE)	○	◐	●	●	◐	●	●	◐
Script/Task Library	○	●	●	●	●	●	●	●
SNMP / Network Device Monitor	○	●	●	●	●	●	●	◐
Policy-Based Configuration	○	●	●	●	●	●	●	●
EDR / AV Integration	○	●	●	●	◐	●	●	◐
Ransomware Detection	○	◐	●	●	◐	●	◐	◐
Linux Agent Support	○	●	●	●	●	●	●	●
macOS Agent Support	○	●	●	●	●	●	●	●
Mobile App (iOS/Android)	○	◐	◐	◐	◐	●	◐	●
AI Autonomous Remediation	○	◐	◐	◐	●	◐	◐	◐
Microsoft 365 Management	○	◐	●	●	◐	◐	◐	◐

CRITICAL MISSING FEATURES

Gap Analysis

FEATURE GAP	PRIORITY	COMPETITORS WITH IT	BUSINESS IMPACT
Real-Time Threshold Alerting CPU/RAM/Disk/Network thresholds with configurable alert rules	CRITICAL	All 7 competitors	Without this, SentinelEdge is reactive (waiting for commands) instead of proactive. MSPs cannot onboard clients who expect automatic issue detection.
Patch Management Automated OS + third-party app patching with approval workflow	CRITICAL	All 7 competitors	Patch management is the #1 RMM purchase driver for MSPs. Its absence makes SentinelEdge non-competitive as a standalone RMM product.
Windows Event Log Monitoring Watch for critical event IDs, login failures, service crashes	CRITICAL	All 7 competitors	Security and compliance use cases require event log visibility. HIPAA, SOC2, and CMMC auditors specifically require log monitoring evidence.

FEATURE GAP	PRIORITY	COMPETITORS WITH IT	BUSINESS IMPACT
Interactive Remote Terminal Real-time bidirectional shell/PS session without script queuing	CRITICAL	All 7 competitors	The current poll-every-30s model adds 0–30s latency per command. Technicians cannot interactively diagnose or remediate issues in real time.
Windows Service Monitor Watch named services; alert and auto-restart on failure	HIGH	All 7 competitors	Production servers running critical services (SQL Server, IIS, backup agents) will have undetected downtime until next heartbeat or human report.
File Transfer (push/pull) Upload scripts/configs to endpoint; pull log files from endpoint	HIGH	All 7 competitors	Technicians must embed large scripts in command payloads. Log collection requires PowerShell that reads and base64-encodes files, creating security surface.
Software Deployment Push MSI/EXE installers to endpoints from central library	HIGH	All 7 competitors	Standard MSP use case: onboarding new devices, deploying security tools, standardizing software stacks across client environments.
Script/Task Library Reusable script library with version control and scheduling	HIGH	All 7 competitors	N-able ships with 650+ scripts; ConnectWise covers 7,000+ app patches. Without a library, SentinelEdge requires technicians to write everything from scratch.

FEATURE GAP	PRIORITY	COMPETITORS WITH IT	BUSINESS IMPACT
Vulnerability Scanning (CVE) Identify installed software with known CVEs; map to patches	MEDIUM	Datto, N-able, Kaseya, CW	Compliance reporting (SOC2/HIPAA/CMMC) requires documented vulnerability posture. MSPs selling security packages cannot demonstrate value without CVE tracking.
Policy-Based Configuration Apply configuration profiles to groups of endpoints automatically	MEDIUM	All 7 competitors	Managing 500+ endpoints with manual per-device commands is operationally impossible. Policies enable standardization at scale.
Linux / macOS Agent Non-Windows endpoint support	MEDIUM	All 7 competitors	MSP clients increasingly have mixed environments. A Windows-only agent covers only part of the endpoint estate.
Process Monitoring Watch named processes; detect unexpected launches or terminations	MEDIUM	All 7 competitors	Security use case: detect unauthorized process execution (crypto miners, RATs). Operational: detect crashed application processes before users report them.
AI Autonomous Remediation LLM-driven self-healing agent (Atera Robin model)	LOW (FUTURE)	Atera (full), 5 others (partial)	Emerging capability. Atera's Robin resolves issues without human intervention. Implementing this would position SentinelEdge as a differentiated, forward-thinking platform.

FEATURE GAP	PRIORITY	COMPETITORS WITH IT	BUSINESS IMPACT
Microsoft 365 Integration Manage M365 users/licenses from agent dashboard	LOW (FUTURE)	Datto (native), N-able (Cloud Commander)	MSP clients are predominantly M365-based. Unified management reduces context-switching for technicians.

TECHNICAL BLUEPRINTS

Implementation Proposals — Phase 1: Core Monitoring

01 · Real-Time Threshold Alerting Engine

CRITICAL

`Go agent` `internal/monitor/` `API: POST /alerts` `~3 weeks`

Add a configurable metrics collection goroutine that samples CPU usage, RAM consumption, disk space per volume, and network interface statistics at a configurable interval (default: 60s). Compare each metric against thresholds stored in a local config section (synced from API on startup). When a threshold is breached, send an alert event to the API with severity, metric name, current value, and threshold. Implement hysteresis to avoid alert storms (don't re-alert until value recovers + grace period).

WHY IT MATTERS

This is the single most impactful gap. It transforms SentinelEdge from a command executor into a proactive monitoring platform — the core value proposition of any RMM.

TECHNICAL APPROACH

- → New `internal/monitor/` package: `metrics.go` (collect), `alerter.go` (evaluate + send), `thresholds.go` (config)
- → Use `github.com/shirou/gopsutil/v3` for cross-platform CPU/RAM/Disk/Network metrics
- → Thresholds pulled from API: `GET /agents/{id}/thresholds` — stored locally, re-synced every hour
- → Alert payload: `{ agent_id, metric, value, threshold, severity, timestamp }`
- → Hysteresis: state machine per metric (Normal → Alerting → Recovering) with configurable recovery window

02 · Windows Event Log Monitor

CRITICAL

`Go agent` `internal/eventlog/` `Windows API: wevtapi.dll` `~2 weeks`

Subscribe to Windows Event Log channels (System, Application, Security, custom) and forward critical events matching configurable event ID rules to the API. This enables detection of login failures (Event 4625), service crashes (7034), disk errors (7, 11), and application crashes — without requiring PowerShell scripts on demand.

WHY IT MATTERS

Security compliance (SOC2 CC7.2, HIPAA §164.312, CMMC AU.2.042) requires continuous log monitoring. Competitors deliver this out of the box; SentinelEdge has no visibility into Windows events today.

TECHNICAL APPROACH

- → Use `golang.org/x/sys/windows` to call `EvtSubscribe` API (push subscription, not polling)
- → Subscribe to: System, Application, Security channels by default; configurable from API
- → Event filter: EventIDs configurable per-tenant (defaults: 4625, 4648, 6008, 7034, 7036, 7040)
- → Batch events in memory (max 50 or 30s) before sending to reduce API calls
- → Include: EventID, Channel, Level, Message (truncated to 500 chars), TimeCreated

03 · Interactive Remote Terminal (WebSocket)

CRITICAL

`Go agent` `internal/terminal/` `WebSocket: gorilla/websocket` `~4 weeks`

Replace the poll-based command model with a WebSocket connection for interactive terminal sessions. When a technician opens a terminal session in the dashboard, the API creates a session token, the agent receives a "open-terminal" command, establishes a WebSocket to the API, spawns a PowerShell or cmd.exe process with I/O pipes, and relays stdin/stdout bidirectionally in real time. The poll-based model remains for non-interactive commands.

TECHNICAL APPROACH

- → New command type `"terminal"` with `{ session_token, shell }` payload
- → Agent dials `wss://saapi.ardepa.site/ws/terminal/{session_token}` with Bearer auth
- → Spawns `powershell.exe -NoExit` using `os/exec`, pipes stdin/stdout/stderr to WS
- → Session timeout: 30 min idle; max session duration: 4 hours (configurable)
- → Audit log: every keystroke session logged server-side (actor + timestamp + session ID)
- → Dashboard UI: xterm.js frontend terminal emulator

> **SECURITY CONSIDERATION**
>
> Terminal sessions must still be gated by the command allowlist for initial shell type. Add separate RBAC permission `terminal.open` — not all technician roles should have interactive shell access.

04 · Patch Management Engine

CRITICAL

`Go agent` `internal/patcher/` `Windows Update API (COM)` `~5 weeks`

Implement a two-phase patch management system: (1) Scan — query Windows Update Agent COM API and installed software for available patches, report to API with CVE mappings; (2) Install — receive approved patch lists from API, download and install patches, report results with reboot requirements. Support approval workflows: patches can be in Pending/Approved/Declined state per-tenant policy.

WHY IT MATTERS

Patch management is the single most commonly purchased RMM feature. Every competitor ships it. Its absence is the most commercially significant gap in SentinelEdge.

TECHNICAL APPROACH

- → Phase 1 (Scan): Use `golang.org/x/sys/windows/registry` + WMI `Win32_QuickFixEngineering` to enumerate installed updates
- → Query Windows Update Agent (WUA) COM via PowerShell wrapper: `Get-WindowsUpdate`
- → Map installed software versions to NVD CVE database (pull from `nvd.nist.gov/feeds` — cached API-side)
- → Phase 2 (Install): Execute approved patches via WUA COM `IUpdateInstaller` interface
- → Report: KBID, status (success/failed/reboot-needed), pre/post install timestamps
- → Schedule: configurable scan every 24h; install window (e.g., Tue 02:00–04:00)

05 · Windows Service & Process Monitor

HIGH

`Go agent` `internal/monitor/services.go` `WMI Win32_Service` `~1.5 weeks`

Monitor the state of named Windows services and critical processes. For services: detect state changes (Running → Stopped), optionally auto-restart, and send alert to API. For processes: watch for unexpected process appearances (security) or disappearances (operational). Configuration is pulled from API per-tenant policy.

- → Poll `Win32_Service` via WMI every 60s; compare state against last known state
- → Config: `{ service_name, expected_state, auto_restart: bool, alert_on_change: bool }`
- → Auto-restart: call `net start {service}` once; mark as "restart attempted" in alert payload
- → Process watch: blocklist and watchlist; alert when blocklisted process spawns (malware detection signal)
- → Process watch uses `gopsutil/v3/process` for cross-platform compatibility

06 · Secure File Transfer

HIGH

`Go agent` `internal/filetransfer/` `Chunked HTTP multipart` `~2 weeks`

Enable two-way file transfer: (1) Push — API sends signed URL for agent to download a file to a specified path; (2) Pull — agent uploads a file from a specified path to the API using multipart upload. All transfers are authenticated, size-limited, path-restricted, and audit-logged.

- → **Push**: new command type `"file-push"` → `{ signed_url, dest_path, sha256, max_size_mb }`
- → **Pull**: new command type `"file-pull"` → `{ src_path, upload_url }`
- → Path restrictions: allowlist of directories where files can be written/read (no system32, no registry hive paths)
- → Max file size: 100MB configurable; SHA256 verified on push before writing
- → Audit log: every transfer logged with actor, path, size, timestamp, direction

07 · Software Deployment Engine

HIGH

`Go agent` `internal/deploy/` `MSI/EXE silent install` `~2.5 weeks`

Deploy software packages (MSI, EXE, ZIP) to endpoints from a central package library stored on the API. The agent downloads the installer, verifies its SHA256, executes it silently, detects installation success by checking the software inventory post-install, and reports the result. Supports both push (immediate) and scheduled (maintenance window) deployment.

- → New command type `"deploy"`: `{ package_id, download_url, sha256, install_args, type }`
- → MSI: `msiexec.exe /i package.msi /quiet /norestart`
- → EXE: configurable silent args (e.g. `/S`, `/silent`) from package manifest
- → Post-install verification: re-run software inventory query for the installed product name
- → Download URL must pass the existing allowlist validation (extending existing security model)

08 · Script & Task Library

HIGH

`API + Dashboard` `Server-side` `Git-backed script storage` `~2 weeks`

Build a centralized script library in the SentinelEdge API and dashboard. Scripts can be categorized (Maintenance, Security, Diagnostics, Compliance), assigned to schedules, and executed against device groups. The library ships with a starter set of commonly needed scripts (disk cleanup, user audit, AV status check) seeded from well-known open-source RMM script repositories.

- → API: `Script` model: id, name, category, language, body, version, sha256, author, created_at
- → Dashboard: Script Library page with search, filter by category, preview, edit, clone
- → Execution: create a `Command` record with `script_id` reference; agent fetches body by ID
- → Scheduling: cron-style schedule per script × device group (e.g. every Sunday at 03:00)
- → SHA256 pre-computed server-side; agent validates before execution (extends existing allowlist model)

09 · Vulnerability Scanning & CVE Mapping

MEDIUM

`API server-side` `NVD CVE feed` `nvd.nist.gov/feeds` `~3 weeks`

The agent already collects software inventory (name + version). Add a server-side CVE matching engine: sync the NVD CVE JSON feed daily, parse CPE (Common Platform Enumeration) identifiers, and match against software inventory records stored per endpoint. Surface vulnerabilities in the dashboard with CVSS scores, affected endpoints, and recommended patches.

- → Agent: no change needed — existing software inventory already captures name + version
- → API: background task syncs `nvd.nist.gov/feeds/json/cve/1.1/` daily into Supabase `cve_entries` table
- → Matching: normalize software name to CPE string, query `cve_entries` for version ranges
- → Dashboard: Vulnerability Overview page — sort by CVSS score, filter by tenant, export to PDF
- → Integration with Patch Management (Impl 04): auto-recommend patch for CVE-matched software

10 · Policy-Based Configuration Engine

MEDIUM

`API + Dashboard + Agent` `Policy evaluation on startup` `~3 weeks`

Allow MSP admins to define configuration policies (alert thresholds, monitoring rules, patch windows, allowed script categories) at the tenant or device-group level. Agents pull their effective policy on startup and on a 1-hour refresh cycle. Policy changes propagate to all devices in a group without manual intervention per device.

- → Policy model: tenant-level defaults → device-group overrides → per-device overrides (inheritance chain)
- → Policy includes: alert thresholds, monitored services list, event IDs to watch, patch approval mode, max concurrent commands
- → Agent: `GET /agents/{id}/policy` on startup and every 3600s; store in memory (not on disk)
- → Dashboard: Policy editor with JSON schema validation, group assignment, effective policy preview
- → Audit: policy changes recorded in audit log (who changed what, when)

11 · Linux & macOS Agent Support

MEDIUM

`Go cross-platform` `GOOS=linux/darwin` `systemd / launchd` `~4 weeks`

Go's cross-compilation makes multi-platform support structurally achievable. Extract all Windows-specific code (WMI inventory, Windows service wrapper, Event Log) behind platform interfaces. Build Linux (systemd) and macOS (launchd) service wrappers. Replace WMI inventory with `gopsutil` for cross-platform metrics.

- → Abstract platform layer: `internal/platform/` with `inventory_windows.go`, `inventory_linux.go`, `inventory_darwin.go` (build tags)
- → Linux inventory: `dmidecode` + `dpkg/rpm` for software; `gopsutil` for metrics
- → macOS inventory: `system_profiler` + `brew list` for software
- → Build pipeline: `GOOS=linux GOARCH=amd64` + `GOOS=darwin GOARCH=arm64` in GitHub Actions
- → kardianos/service already supports Windows/Linux/macOS — no service wrapper rewrite needed

12 · AI-Assisted Autonomous Remediation

LOW — FUTURE

`Claude Sonnet API` `Agent-side execution` `Tool-calling pattern` `~6 weeks`

Inspired by Atera's Robin, build an AI remediation loop: when an alert fires, instead of only notifying technicians, optionally invoke an AI agent (Claude Sonnet) with the alert context, system metrics, and recent event logs. The AI can call approved diagnostic tools (commands from the allowlist), read results, and attempt remediation. All AI actions are gated by the allowlist and require technician approval for destructive steps.

- → AI agent receives: alert type, current metrics snapshot, last 50 event log entries, installed services list
- → AI can call: approved diagnostic PS scripts (disk analysis, process list, network stats)
- → Remediation actions requiring approval: service restarts, patch installs, process termination
- → Technician approval via Slack/Teams webhook or dashboard notification before execution
- → Full audit trail: AI reasoning steps, tools called, results, final action taken

IMPLEMENTATION TIMELINE

Phased Roadmap

The roadmap is structured in four phases based on business impact, technical dependency, and implementation complexity. Phases 1 and 2 close the critical competitive parity gaps. Phases 3 and 4 build differentiation.

PHASE 1 · Q2 2026 (MONTHS 1–3)

Monitoring Foundation — Achieve Competitive Parity on Core Features

Impl 01 Real-Time Threshold Alerting (CPU/RAM/Disk/Network) · 3 weeks
Impl 02 Windows Event Log Monitor · 2 weeks
Impl 05 Windows Service & Process Monitor · 1.5 weeks
Security Resolve F-01 (TLS pinning) + F-05 (command nonces)

Deliverable: SentinelEdge can proactively alert on endpoint health issues without technician polling. Required for any serious MSP evaluation.

PHASE 2 · Q3 2026 (MONTHS 4–6)

Operational Capabilities — Enable Day-to-Day MSP Workflows

Impl 03 Interactive Remote Terminal (WebSocket) · 4 weeks
Impl 06 Secure File Transfer (push/pull) · 2 weeks
Impl 07 Software Deployment Engine (MSI/EXE) · 2.5 weeks
Impl 08 Script & Task Library (API/Dashboard) · 2 weeks

Deliverable: Technicians can perform full endpoint management workflows from the SentinelEdge dashboard without leaving the platform.

PHASE 3 · Q4 2026 (MONTHS 7–9)

Security & Compliance — Position SentinelEdge for Security-Conscious MSPs

Impl 04 Patch Management Engine (OS + software) · 5 weeks
Impl 09 Vulnerability Scanning & CVE Mapping · 3 weeks
Impl 10 Policy-Based Configuration Engine · 3 weeks
Impl 11 Linux & macOS Agent Support · 4 weeks

Deliverable: SentinelEdge can support MSPs with SOC2/HIPAA/CMMC compliance requirements. Cross-platform coverage enables mixed-OS client environments.

PHASE 4 · Q1 2027 (MONTHS 10–12)

Differentiation — AI, Mobile, and M365 Integration

Impl 12 AI Autonomous Remediation (Claude-powered) · 6 weeks
Future Mobile app (React Native) for iOS/Android · 6 weeks
Future Microsoft Graph / M365 tenant management · 4 weeks
Future SNMP network device monitoring · 3 weeks

Deliverable: SentinelEdge is differentiated from incumbents with AI-first remediation — a capability only Atera delivers today, giving SentinelEdge a unique selling proposition.

FINAL ASSESSMENT

Conclusions & Recommendations

SentinelEdge Agent has a technically sound, security-forward foundation that compares favorably to incumbents in credential management (Vaultwarden), command security (deny-by-default allowlist), and update integrity (SHA256 verification). However, the feature set represents approximately **~20% of the baseline capability** expected of a commercially viable RMM agent in 2026.

> **COMPETITIVE ADVANTAGE ALREADY BUILT**
>
> SentinelEdge's deny-by-default command allowlist with SHA256 hash registry is **more security-rigorous than any competitor's default configuration**. This is a genuine differentiator that can be marketed to security-conscious MSPs and compliance-driven verticals (healthcare, finance, government contractors).

Top 4 Actions to Take Immediately

→ **Start Impl 01 (Alerting Engine)** — this single feature transforms SentinelEdge from a command runner to a monitoring platform. Highest ROI per development hour.

→ **Start Impl 02 (Event Log Monitor)** — enables compliance use cases. Can be developed in parallel with alerting. Unlocks SOC2/HIPAA/CMMC customer conversations.

→ **Fix F-01 CRITICAL (TLS Pinning)** — outstanding security debt from the April 2026 audit. Should not ship new features until critical security findings are resolved.

→ **Design Impl 03 (Remote Terminal)** — resolves the 30-second latency ceiling that prevents interactive troubleshooting. Required before Phase 2 work begins.

Unique Opportunity in the Market

Atera launched Robin (AI autonomous remediation) in March 2026 and it is still early-stage. No other RMM has a mature, production-ready AI remediation agent. Implementing Impl 12 (AI Autonomous Remediation using Claude Sonnet) before competitors mature their offerings gives SentinelEdge the opportunity to be the **"AI-native RMM" positioning** — a significant market differentiation point. Given that the existing team already integrates Claude AI (see Finanzas Ardepa), this is a realistic near-term capability.

> **API-SIDE ISSUE REQUIRING PARALLEL ACTION**
>
> The SentinelEdge API stores `Agent.token` in plaintext in Supabase. The `hash_token()` function exists in `core/security.py` but is not wired up. This must be resolved before any public release — a compromised Supabase database would expose all agent tokens across all tenants.
